

Matching and Perfect Matching Polytopes

$$G = (V, E)$$

$$P_{PM} \triangleq \text{conv} \{ x^M \in \mathbb{R}^E : M \text{ is a perfect matching} \}$$

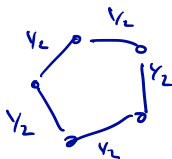
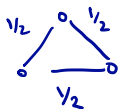
$$P_M \triangleq \text{conv} \{ x^M \in \mathbb{R}^E : M \text{ is a matching} \}$$

Recall that for bipartite:

$$x(\delta(v)) \leq 1 \quad \text{for } P_M$$

$$= 1$$

$$P_{PM} \leftarrow$$



Odd Subset Constraints

To rule out:



$$x(E[U]) \leq \lfloor \frac{1}{2} |U| \rfloor \quad \forall U \subseteq V, |U| \text{ odd}$$

$\hookrightarrow$ set of edges in the subgraph defined by the nodes in U .

$$U = \text{triangle graph}, \quad x(E[U]) = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2} > 1.$$

$$\underline{\text{Thm}}: \left\{ x(\delta(v)) \leq 1, \quad x \geq 0, \quad x(E[U]) \leq \lfloor \frac{1}{2} |U| \rfloor \quad \forall |U| \text{ odd} \right\}$$

Q_M

$$Q_M = P_M.$$

[Edmonds]

Perfect Matching Polytope

$$Q'_{PM} = \left\{ x \geq 0, \quad \cancel{x(\delta(v)) = 1}, \quad \underline{x(E(U)) = \lfloor \frac{1}{2}|U| \rfloor} \right. \\ \left. |U| \text{ odd} \right\}$$

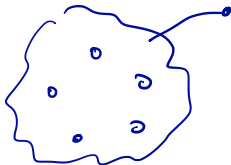
Instead of working with Q'_{PM} , we show

$$\text{that: } Q_{PM} = \left\{ \underline{x \geq 0}, \quad \underline{x(\delta(v)) = 1}, \quad \underline{x(\delta(U)) \geq 1}, \quad \forall U \subseteq V, |U| \text{ odd} \right\}$$

Thm: $Q_{PM} = P_{PM}$ (Edmonds).

$$[\text{Next lecture: } \underline{Q_{PM} = P_{PM}} \Rightarrow \underline{Q'_{PM} = P_{PM} = Q_{PM}}]$$

pf: $P_{PM} \overset{\checkmark}{\subseteq} Q_{PM}$



Proof of Thm — 1/5

If $Q_{PM} \subsetneq P_{PM} \Rightarrow \exists x \in Q_{PM}$ that is a vertex
that is not in P_{PM} .

Let G be the smallest graph, i.e. the graph minimizing $|V| + |E|$, that exhibits above counterexample.

Back to our vertex $x \in Q_{PM} \setminus P_{PM}$,

Claim: $0 < x_e < 1$, o.w. we contradict minimality of G
b/c we could delete edge (if $x_e = 0$) or vertex ($x_e = 1$)

Implies: $\deg(v) \geq 2 \quad \forall v \in V$, and $|E| \geq |V|$

If $|E| = |V| \Rightarrow$ each node has degree = 2, also

all cycles are even length.

Ex: Thus, x is not an ext pt ~~*~~.



Proof of Thm — 2/5

Thus, have left to consider:

$\deg(v) \geq 2, \forall v \in V$, and $|E| > |V|$.

2 is a vertex. $\Rightarrow$ satisfying $|E|$ tight constraints

" $x \geq 0$ " — none are tight

" $x(\delta(u)) = 1$ " — giving $|V|$ tight & indep. constraints

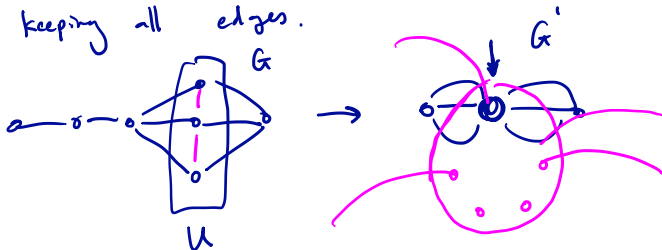
$\Rightarrow \exists u, |u|$ odd and $3 \leq |u| \leq |V| - 3$

with $x(\delta(u)) = 1$.

Proof of Thm — 3/5

Make 2 new graphs: $G' = G/U$, $G'' = G/\bar{U}$

G/U , $G/\bar{U}$ are created by contracting nodes in U , $\bar{U}$ resp. by to a single node and keeping all edges.



x naturally defines x' and x'' by removing coeff. corresponding to internal edges.

Now: $x' \geq 0$ ✓ $x'(\delta(v)) = 1 \forall v \in G'$ ✓ b/c $x(\delta(U)) = 1$ by assumption
 $x'(\delta(u')) \geq 1 \forall u' \in V'$ odd ✓ b/c it is inherited from x in G
 b/c we did not change the parity of any subst. $\Rightarrow x' \in Q'_{PM} = P'_{PM}$
 $x'' \in Q''_{PM} = P''_{PM}$

Proof of Thm — 4/5

x rational $\Rightarrow x', x''$ rational, hence
by repeating perfect matchings if necessary:

$\exists M'_1, M'_2, \dots, M'_k$, p.m. in $G' = G/\underline{u}$

$\exists M''_1, M''_2, \dots, M''_k$ p.m. in $G'' = G/\overline{u}$

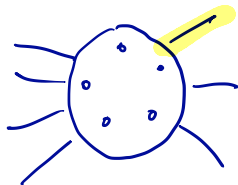
$$\text{s.t. } (*) \quad \left[\begin{array}{l} x' = \frac{1}{k} \sum x^{M'_i} \\ x'' = \frac{1}{k} \sum x^{M''_i} \end{array} \right]$$

Recall that edges in $\delta(u)$ were not deleted when forming G'
similarly for G'' .

$$\Rightarrow \forall e \in \delta(u), \quad \underline{k x'(e) = k \cdot x(e) = k x''(e)}$$

Hence: $\forall e \in \delta(u)$ the number of p.m. M'_i with $e \in M'_i$
equals the number of matchings M''_i with $e \in M''_i$.

Proof of Thm — 5/5



$\delta(u)$



M_i'

Each matching M_i', M_i'' contains exactly one edge $e \in \delta(u)$

By renumbering if necessary, pair up $M_i' \leq M_i''$ so that for all i , $M_i' \leq M_i''$ contain the same edge $e \in \delta(u)$.

$$x = \begin{pmatrix} \overline{u-u} & \overline{u-u} & \overline{u-u} \end{pmatrix}$$

$$x' = \begin{pmatrix} \times & \dots & \dots \end{pmatrix}$$

$$x'' = \begin{pmatrix} \dots & \times & \dots \end{pmatrix}$$

$$M_i = M_i' \cup M_i'' : M_i \text{ is a PM}$$

Then: $x = \frac{1}{k} \sum x^{M_i}$

Next:

Use PM result to show $Q_M \subseteq P_M$

Immediately implies: $Q'_{PM} = Q_{PM}$.